

LETTERS TO THE EDITOR

Under this heading brief reports of important discoveries in physics may be published. These reports should not exceed 600 words and, for any issue, should be submitted not later than six weeks previous to the first day of the month of issue. No proof will be sent to the authors.

Absorption Spectrum of Free CH₃ and CD₃ Radicals¹

Numerous attempts to obtain the absorption spectrum of the free CH₃ radical have been made in the spectral region 2200–9000 Å. It is possible that the failure of these attempts is due to the fact that the electronic transitions expected in this region are forbidden and therefore far too weak for detection, or to the fact that the upper state is unstable and therefore the absorption is continuous and consequently not easily detected. If CH₃ is planar, the first reason is very likely to apply (see Walsh 1953). It was therefore considered worth while to search for an absorption spectrum in the vacuum ultraviolet where strong allowed absorptions of the Rydberg type are expected.

The radicals were produced by the flash photolysis technique and the absorption spectrum was investigated with the 3 meter vacuum grating spectrograph of this laboratory. Using Hg(CH₃)₂ as the parent compound at a pressure of 0.04 mm. in an absorption tube of 50 cm. length, four narrow groups of diffuse absorption bands in the region 1510–1300 Å were found when absorption spectra were taken between 50 and 250 μsec. after the photolysis flash. Similar but slightly shifted band groups were observed in the flash photolysis of Hg(CD₃)₂. When, under the optimum conditions, the study of the spectrum was extended to the longer wavelength region, a further band was observed near 2160 Å in Hg(CH₃)₂ and near 2140 Å in Hg(CD₃)₂. In this region, the absorption of the parent compound is not as troublesome as in the 1500 Å region and therefore the 2150 Å bands can be obtained with considerable intensity by increasing the pressure. They are shown in Fig. 1. At the increased pressure, in Hg(CD₃)₂ two additional much weaker bands appear, one of which is visible in Fig. 1.

TABLE I
ABSORPTION BANDS OF CH₃ AND CD₃

n	CH ₃		CD ₃	
	ν _{obs} (cm ⁻¹)	ν _{calc} (cm ⁻¹)	ν _{obs} (cm ⁻¹)	ν _{calc} (cm ⁻¹)
	{ 46205 46333 v.w.		{ 46626.2 Q head 46675 R peak 47253 Q head 47304 R peak	
3	{ 66536 v.s. 66717 s. 66811 s.	66536	{ 66471 v.s. 66730 s.	66461
4	{ 72168 s. 72329 s.	72250	{ 72177 m. 72300 v.w.	72175
5	74852 m.	74852	74771 m.	74777
6	76252 m.	76252	76178 w.	76177
7	77088 w.	77090		

In Table I all the observed bands are listed. These bands occur only immediately (within less than 250 μsec.) after the photolysis flash which lasts about 50 μsec. They do not occur in undecomposed Hg(CH₃)₂ nor in the residue, say two minutes after photolysis. They must therefore be due to a molecule of short life that is produced by the photodecomposition. Since, from photochemical evidence (see Steacie 1954), it is known that Hg(CH₃)₂ splits off one or two CH₃ radicals in the primary act of photodecomposition, it appears probable that the observed spectra are due to free CH₃ and CD₃, respectively.

Further strong evidence for free methyl as the carrier of these spectra comes from the observation of the same bands (to the extent that they are not hidden by parent absorption) in the photolysis of CH₃CHO, (CH₃)₂N₂, (CH₃)₂CO, CH₃I, and CH₃Br. For all these molecules there is photochemical evidence that free CH₃ radicals are produced.

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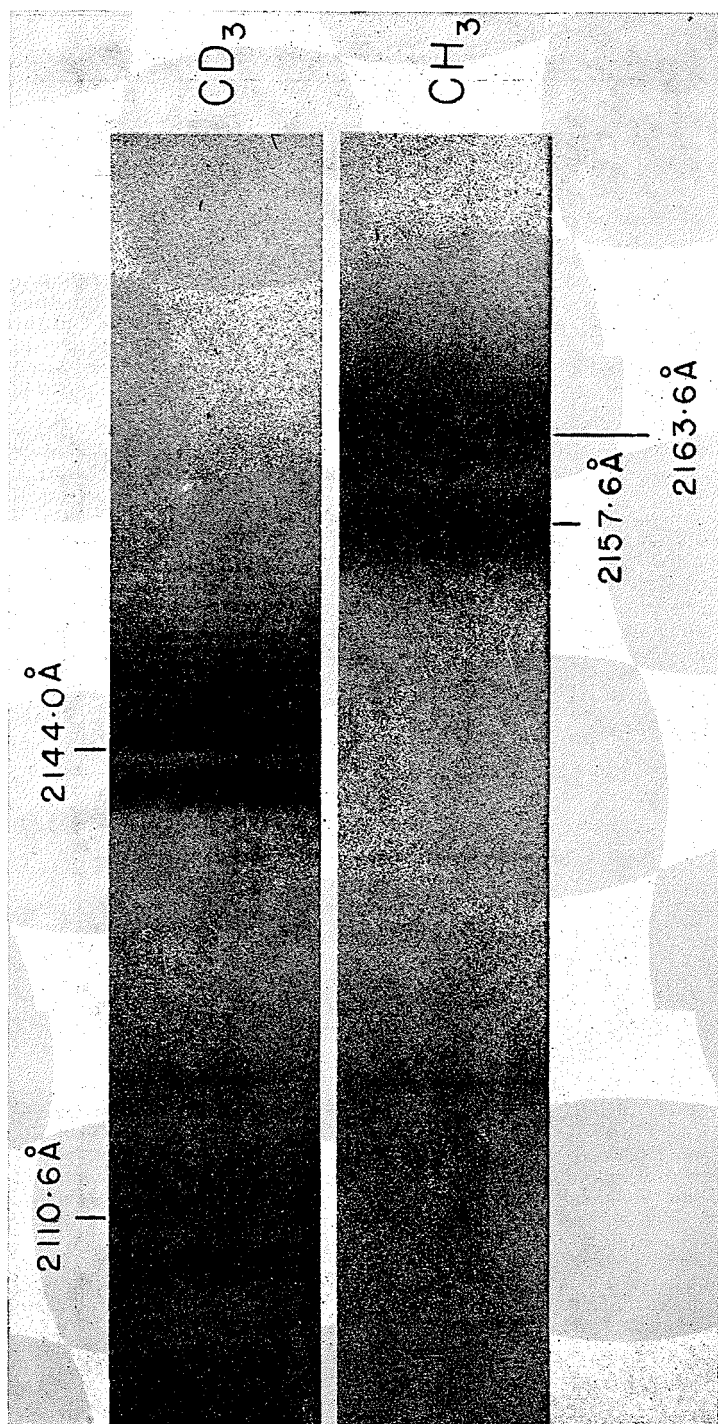


FIG. 1. Absorption spectra of CH_3 and CD_3 . (The first wave length at the top should be $2115.6 \text{ \AA}$ not $2110.6 \text{ \AA}$.)

The principal bands in the vacuum region form clear Rydberg series which can be represented by the formulae

$$\text{CH}_3: \nu = 79380 - \frac{R}{(n - 0.077)^2}, \quad n = 3, 4, 5, 6, 7, \dots$$

$$\text{CD}_3: \nu = 79305 - \frac{R}{(n - 0.077)^2}, \quad n = 3, 4, 5, 6, \dots$$

The wave numbers calculated from these formulae are included in Table I. The constant terms in these formulae represent the ionization potentials of CH_3 and CD_3 . Converted to electron volts they are 9.840 and 9.832 ± 0.002 ev. The excellent agreement with the value obtained by various authors (Lossing, Ingold, and Henderson 1954; Langer, Hipple, and Stevenson 1954) from mass spectrometric measurements, namely 9.90 ± 0.1 ev., makes it practically certain that the observed bands are due to free CH_3 and CD_3 .

Further conclusive proof that the 2150 Å bands shown in Fig. 1 are due to CH_3 and CD_3 has been obtained by a study of these bands in the flash photolysis of isotopic mixtures of acetone containing 25, 50, and 75% deuterium. In these mixtures two additional bands appear in the space between the CH_3 and CD_3 bands, clearly belonging to CHD_2 and CD_2H . Thus there can be no doubt that the molecule responsible for the spectrum under discussion has three H atoms. The coarseness of the rotational structure of the 2140 Å bands of CD_3 (Fig. 1) excludes the presence of more than one C atom in the molecule. Therefore, one can safely and unambiguously conclude that the spectrum is due to the free methyl radical.

With one exception, the CD_3 band at 2140 Å, all observed bands are diffuse without any indication of fine structure. This diffuseness is clearly due to predissociation which, in the case of the 2140 Å CD_3 band, is apparently slowed down by a barrier which is less easily penetrated in CD_3 than in CH_3 . The one resolved branch of the 2140 Å band shows a slight intensity alternation. Assuming the band to be a $\parallel$ band of a planar CD_3 molecule, a slight intensity alternation is expected since the subband with $K = 0$ would have a 10:1 intensity alternation while the other subbands have none, leaving a small alternation in the superposed subbands. The same would apply for a non-planar CD_3 molecule if the inversion doubling is large enough to give rise to separate bands for the two components, that is, if the deviation from the planar configuration is small; but no intensity alternation is expected if the inversion doubling is small, i.e., if the molecule is far from being planar.

The conclusion that the CH_3 molecule in its ground state is planar or nearly planar is in agreement with the predictions of molecular orbital theory (see Walsh 1953) and at the same time accounts in a satisfactory way on the basis of the Franck-Condon principle for the observation of a simple Rydberg series of bands rather than of complex band systems as exhibited by NH_3 in the region below 1500 Å.

With the CH_3 spectrum here observed it will be possible to search more readily for improved conditions of increased concentration and in this way it should be possible to find the expected forbidden transition in the visible region. If this transition occurs above about 2600 Å, it could not be predissociated and would therefore give more precise information on the geometrical structure of CH_3 .

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